

# Chien-Yu Lin

Research Scientist  
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## Research Interests

I'm passionate about **making machine learning efficient**. My research spans a **wide range of ML workloads**, including LLMs, NeRFs, GNNs and CNNs, and covers **multiple domains**, such as efficient training and inference algorithms, GPU kernels and accelerator designs. Moving forward, I aim to expand my cross-stack research to develop highly efficient systems for multi-modal models, explore novel model architectures beyond transformers, and investigate the robustness for efficient techniques.

## Education

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|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sep 2020 - Jun 2025 | Ph.D., Computer Science and Engineering<br>University of Washington, USA<br>Thesis: Toward Efficient Machine Learning Systems with Sampling and Compression<br>Advisor: Prof. Luis Ceze           |
| Sep 2018 - Jun 2020 | M.Sc., Computer Science and Engineering<br>University of Washington, USA                                                                                                                          |
| Sep 2015 - Jun 2017 | M.Sc., Electronics Engineering<br>National Yang Ming Chiao Tung University, Taiwan<br>Thesis: A Dual-Sparsity Accelerator for Sparse Convolutional Neural Networks<br>Advisor: Prof. Bo-Cheng Lai |
| Jan 2015 - Jun 2015 | Exchanged student<br>Koc University, Istanbul, Turkey                                                                                                                                             |
| Sep 2011 - Jan 2015 | B.Sc., Electronics Engineering<br>Minor in Computer Science<br>National Yang Ming Chiao Tung University, Taiwan                                                                                   |

## Work Experience

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| Aug 2025 - Present   | <b>Research Scientist</b><br>Meta MSL (FAIR group), Menlo Park, USA <ul style="list-style-type: none"><li>Efficient machine learning systems.</li><li>Working with Dr. Carole-Jean Wu.</li></ul>                                                                                                                                                        |
| Mar 2023 - Jun 2023  | <b>Machine Learning Research Intern</b>                                                                                                                                                                                                                                                                                                                 |
| Oct 2021 - Sep 2022  | AI/ML org., Apple Inc, Seattle, USA <ul style="list-style-type: none"><li>1st time hosts: Anish Prabhu, Carlo Del Mundo, Mohammad Rastegari.</li><li>2nd time hosts: Thomas Merth and Anurag Rajan.</li><li>Research on model compression and efficient 3D rendering algorithms.</li><li>Published one <b>ECCV</b> and one <b>WAVC</b> paper.</li></ul> |
| Jan 2018 - Aug 2018  | <b>Algorithm Engineer Intern</b><br>Ambarella Inc, Santa Clara, USA <ul style="list-style-type: none"><li>Developed efficient CNN models for self-driving cars.</li></ul>                                                                                                                                                                               |
| July 2014 - Aug 2014 | <b>Compiler Engineer Intern</b><br>Marvell, Hsinchu, Taiwan <ul style="list-style-type: none"><li>Built a verification tool for an advanced in-house C++ compiler.</li></ul>                                                                                                                                                                            |

## Publications

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(\* indicating equal contribution)

- [1] TeleRAG: Efficient Retrieval-Augmented Generation Inference with Lookahead Retrieval.  
**Chien-Yu Lin\***, Keisuke Kamahori\*, Yiyu Liu\*, Xiaoxiang Shi, Madhav Kashyap, Rulin Shao, Yile Gu, Zihao Ye, Kan Zhu, Rohan Kadekodi, Arvind Krishnamurthy, Stephanie Wang, Luis Ceze, Baris Kasikci.  
In The Ninth Annual Conference on Machine Learning and Systems (**MLSys**) 2026.
- [2] UniQL: Unified Quantization and Low-rank Compression for Adaptive Edge LLMs.  
Hung-Yueh Chiang, Chi-Chih Chang, Yu-Chen Lu, **Chien-Yu Lin**, Kai-Chiang Wu, Mohamed S. Abdelfattah, Diana Marculescu.  
In The Fourteenth International Conference on Learning Representations (**ICLR**), 2026.
- [3] Composer: A Search Framework for Hybrid Neural Architecture Design.  
Bilge Acun, Prasoon Sinha, Newsha Ardalani, Sangmin Bae, Alicia Golden, **Chien-Yu Lin**, Meghana Madhyastha, Fei Sun, Neeraja J. Yadwadkar, Carole-Jean Wu.  
In The Fourteenth International Conference on Learning Representations (**ICLR**), 2026.
- [4] NanoFlow: Towards Optimal Large Language Model Serving Throughput.  
Kan Zhu, Yilong Zhao, Liangyu Zhao, Gefei Zuo, Yile Gu, Dedong Xie, Yufei Gao, Qinyu Xu, Tian Tang, Zihao Ye, Keisuke Kamahori, **Chien-Yu Lin**, Stephanie Wang, Arvind Krishnamurthy, Baris Kasikci.  
In The 19th USENIX Symposium on Operating Systems Design and Implementation (**OSDI**), 2025.
- [5] Palu: Compressing KV Cache via Low-Rank Projection.  
Chi-Chih Chang\*, Wei-Cheng Lin\*, **Chien-Yu Lin\***, Yu-Fang Hu, Pei-Shuo Wang, Chong-Yan Chen, Ning-Chi Huang, Luis Ceze, Mohamed S. Abdelfattah, Kai-Chiang Wu.  
In The Thirteenth International Conference on Learning Representations (**ICLR**), 2025  
**First low-rank based KV-Cache compression for LLMs** (Concurrent to DeepSeek MLA).
- [6] Atom: Low-bit Quantization for Efficient and Accurate LLM Serving.  
Yilong Zhao, **Chien-Yu Lin**, Kan Zhu, Zihao Ye, Lequn Chen, Size Zheng, Luis Ceze, Arvind Krishnamurthy, Tianqi Chen, Baris Kasikci.  
In The Seventh Conference on Machine Learning and Systems (**MLSys**), 2024 (Accept rate 22%)  
**One of the earliest 4-bit weight activation quantization for LLMs..**
- [7] FastSR-NeRF: Improving NeRF Efficiency on Consumer Devices with A Simple Super-Resolution Pipeline.  
**Chien-Yu Lin**, Qichen Fu, Thomas Merth, Karren Yang, Anurag Ranjan.  
In The IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**), 2024.  
**Oral (Top 2.6%).**
- [8] SPIN: An Empirical Evaluation on Sharing Parameters of Isotropic Networks.  
**Chien-Yu Lin\***, Anish Prabhu\*, Thomas Merth, Sachin Mehta, Anurag Ranjan, Maxwell Horton, and Mohammad Rastegari.  
In The European Conference on Computer Vision (**ECCV**), 2022.
- [9] Enhancing Utilization of SIMD-Like Accelerator for Sparse Convolutional Neural Networks.  
Bo-Cheng Lai, Jyun-Wei Pan, and **Chien-Yu Lin**.  
In IEEE Transactions on Very Large Scale Integration Systems (**TVLSI**), Feb. 2019.
- [10] Supporting Compressed-Sparse Activations and Weights on SIMD-like Accelerator for Sparse Convolutional Neural Networks.  
**Chien-Yu Lin** and Bo-Cheng Lai.  
In the 23rd Asia and South Pacific Design Automation Conference (**ASP-DAC**), 2018.

## Technical Reports

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- [1] Hybrid Architectures for Language Models: Systematic Analysis and Design Insights.  
Sangmin Bae, Bilge Acun, Haroun Habeeb, Seungyeon Kim, **Chien-Yu Lin**, Liang Luo, Junjie Wang, Carole-Jean Wu.  
In submission to TMLR 2026. ArXiv:2510.04800.

- [2] xKV: Cross-Layer SVD for KV-Cache Compression.  
Chi-Chih Chang, **Chien-Yu Lin**, Wei-Cheng Lin, Yash Akhauri, Hung-Yueh Chiang, Xilai Dai, Huiqiang Jiang, Yucheng Li, Kai-Chiang Wu, Luis Ceze, Mohamed S. Abdelfattah.  
In submission to ICML 2026. ArXiv:2503.18893, 2025.
- [3] Efficient Encoder-Decoder Transformer Decoding for Decomposable Tasks.  
Bo-Ru Lu, Nikita Haduong, **Chien-Yu Lin**, Hao Cheng, Noah A. Smith, Mari Ostendorf.  
ArXiv:2403.13112, 2024.
- [4] Accelerating SpMM Kernel with Cache-First Edge Sampling for Graph Neural Networks.  
**Chien-Yu Lin**, Liang Luo, and Luis Ceze.  
ArXiv:2104.10716, 2021.

## Teaching Experience

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|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| Fall 2024   | <b>Systems for Machine Learning, CSE 599K, UW</b><br>Guest Lecturer and Teaching Assistant <ul style="list-style-type: none"> <li>• With Prof. Arvind Krishnamurthy.</li> <li>• <b>Taught three lectures</b> on LLM performance optimizations and ML hardware.</li> <li>• Designed an assignment on attention performance analysis.</li> <li>• Link: <a href="https://courses.cs.washington.edu/courses/cse599k/24au/">https://courses.cs.washington.edu/courses/cse599k/24au/</a></li> </ul> |
| Spring 2024 | <b>High-Performance Scientific Computing, Amath 483/583 A, UW</b><br>Teaching Assistant <ul style="list-style-type: none"> <li>• With Prof. Kenneth Roche.</li> <li>• Primary parallel computing course in UW.</li> <li>• Topics cover pthreads, multi-process, MPI, and CUDA.</li> </ul>                                                                                                                                                                                                     |
| Spring 2022 | <b>Computer Architecture II, CSE 470, UW</b><br>Teaching Assistant <ul style="list-style-type: none"> <li>• With Prof. Luis Ceze.</li> </ul>                                                                                                                                                                                                                                                                                                                                                  |
| Fall 2016   | <b>Computer Architecture (Grad Level), EE, NYCU</b>                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Fall 2015   | <b>Computer Organization (Undergrad Level), EE, NYCU</b>                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Spring 2015 | Teaching Assistant <ul style="list-style-type: none"> <li>• With Prof. Bo-Cheng Lai.</li> <li>• Designed several new course projects. Topics included acceleration of image processing and dense/sparse neural networks.</li> <li>• Tools involved RISC-V toolchain, Multi2Sim and CUDA programming.</li> </ul>                                                                                                                                                                               |

## Service

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| 2026        | Reviewer, ECCV                                                                                                                                                                                                                                                                         |
| 2026        | External Reviewer, MLSys                                                                                                                                                                                                                                                               |
| 2026        | Reviewer, ICLR                                                                                                                                                                                                                                                                         |
| 2023 - 2025 | Lab seminar organizer, SAMPL Lab, UW <ul style="list-style-type: none"> <li>• Revived lab activities from COVID hiatus to 30+ weekly attendees.</li> <li>• Events link: <a href="https://sAMPL.cs.washington.edu/talks.html">https://sAMPL.cs.washington.edu/talks.html</a></li> </ul> |
| 2024 - 2025 | PhD admission committee area chair, UW CSE                                                                                                                                                                                                                                             |
| 2024        | Reviewer, 3DV                                                                                                                                                                                                                                                                          |
| 2021        | PhD admission committee, UW CSE                                                                                                                                                                                                                                                        |
| 2020        | Artifact evaluation committee, ASPLOS                                                                                                                                                                                                                                                  |
| 2020        | Prospective student committee chairs, CSE, UW                                                                                                                                                                                                                                          |
| 2013 - 2014 | Student system administrator, EE, NYCU                                                                                                                                                                                                                                                 |

## Awards

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|------|---------------------------------------------------------------------------------|
| 2024 | MLSys student travel grant.                                                     |
| 2024 | Oral award (top 2.6%), WACV 2024.                                               |
| 2014 | Outstanding student, System and architecture talent incubation program, Taiwan. |

## Mentoring

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I find great joy in helping junior students develop skills and achieve their goals. I am fortunate to mentor the following brilliant students.

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|-------------------------|--------------------------------------------------------------|
| Fall 2024 - Spring 2025 | Yiyu Liu (SJTU), now a CS PhD student in Harvard University. |
| Spring 2024 - 2025      | Chi-Chih Chang (NYCU), now an ECE PhD student in Cornell.    |
| Summer - Fall 2023      | Yilong Zhao (SJTU), now an EECS PhD student in UC Berkeley.  |
| Spring 2017             | Jyun-Wei Pan (NYCU), now an engineer at MediaTek.            |

## Patents

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- [P1] Apparatus and Method of Using Dual Indexing in Input Neurons and Corresponding Weights of Sparse Neural Network.  
**Chien-Yu Lin**, and Bo-Cheng Lai.  
US Patent Application 15/594,667, 2018.

## Invited Talks

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|-------------|-----------------------------------------------------------------------------------------|
| April. 2025 | Toward efficient machine learning systems with sampling and compression, Amazon AWS AI. |
| Mar. 2025   | Toward efficient machine learning systems with sampling and compression, Meta FAIR.     |
| Mar. 2025   | Efficient 3D neural rendering and LLM inference, Apple MLR.                             |
| Jan. 2025   | Efficient RAG inference with lookahead retrieval, at NYCU, Taiwan.                      |
| Jan. 2025   | LLM quantization and KV-Cache compression, at NTU, Taiwan.                              |
| Nov. 2024   | KV-Cache compression with low-rank projection, at UW CSE research day, Seattle.         |
| May. 2024   | Low-bit quantization for LLMs, at MLSys, Santa Clara.                                   |
| Jan. 2024   | Low-bit quantization for LLMs, at NCKU, Taiwan.                                         |
| Jan. 2024   | Fast NeRF with super resolution, at WACV.                                               |
| Nov. 2023   | Low-bit quantization for large language models, at UW CSE research day, Seattle.        |
| Jan. 2018   | Accelerator design for sparse CNNs, at ASP-DAC, South Korea.                            |
| Jun. 2016   | A Survey of CNN Accelerators, at MediaTek, Hsinchu.                                     |

## Mountain Experience

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In addition to research, I have a strong passion for exploring nature, particularly through **mountaineering**, **backcountry skiing** and **bike packing**. I frequently lead groups on mountain expeditions and have had many successful summits and ski descents on challenging peaks in the Pacific Northwest (PNW), including Mt. Rainier (14,410 ft), Mt. Shasta (14,179 ft), and Mt. Hood (11,249 ft). In summer 2025, I completed a **1000 miles bike trip** from Seattle, WA, to Menlo Park, CA, following Hwy 101 and pacific coast. These experiences have taught me invaluable lessons in team leadership, risk management, and resilience - skills that I apply in my professional work :).